

My research focuses on building efficient, next-generation computer systems for machine learning workloads. This aspiration was forged during my time at AWS Shanghai and NYU, where I observed a growing disparity between rapid algorithmic innovation and the practical constraints of production deployment. To bridge this gap, I aim to explore a full-stack approach that combines three key directions: developing new abstractions for emerging algorithms, implementing hardware-aware optimizations, and redesigning end-to-end pipelines for holistic resource utilization. Backed by my research and industry experiences that have led to publications at top-tier venues such as **SOSP'23**, **SIGMOD'25**, **PPoPP'25**, and **ICLR'25**, I am eager to pursue a Ph.D. at MIT to further advance the scalability and robustness of future AI infrastructure.

Research Experience at SUSTech & AWS Shanghai AI Lab

I had both my undergraduate and master's studies at DBGroup@SUSTech, advised by **Prof. Bo Tang** and **Prof. Xiao Yan**. My foundational experience was forged during my internship at AWS Shanghai, working with **Dr. Minjie Wang**. I started by working on optimizing **GNN training across diverse scenarios**. On the algorithmic side, I co-authored MuseGNN [2] (**ICLR 2025**), designing a new GNN architecture that scales to large graphs. This algorithm perspective guided my primary focus on systems research. During the exploration, I observed a surge in new GNN sampling algorithms that existing systems could not support, leading to ad-hoc implementation and inefficient sampling. As such, I identified a critical need for a new system abstraction to enable **efficient graph sampling on GPU**. After studying 15 popular graph sampling algorithms, I modeled their sampling processes using unified matrix operations. Under this new abstraction, I proposed IR-based GPU kernel fusion to eliminate redundant data materialization and irregular subgraph batching to improve GPU utilization. These combined efforts yielded up to 20x speedup in the sampling process, and this work, gSampler [1], was accepted by **SOSP 2023**.

Extending my focus to the **distributed training scenario**, I led a project to tackle communication bottlenecks in collaboration with **Prof. James Cheng** at CUHK. Through preliminary exploration, I observed that no single GNN parallelism performs optimally under all configurations. By viewing GNN computation as sparse matrix multiplication, I summarized all possible parallelisms cutting along different dimensions and designed a framework to adaptively determine the optimal strategy. To implement a unified execution engine for parallelism selection, I propose to automatically insert GPU communication operators at model layer barriers and designed a profiling-based cost model to precisely navigate the complex trade-offs. This auto-parallelism enabled an over 2x speedup compared to always using a static, single-strategy approach. These efforts led to my co-first-author publication, APT [3], at **PPoPP 2025**.

To further enable **training on billion-scale graphs exceeding CPU memory**, I then led a project to address the I/O bottleneck. Contrary to popular approaches that suffer from either disk read amplification or degraded model accuracy, I proposed an **offline sampling scheme**, where all mini-batches are prepared before model computation. This new scheme allows us to continuously store node features of a batch on disk for better access locality, all while preserving the original training algorithm, thus ensuring no model accuracy loss. This system, together with hierarchical feature caching and elaborate overlapping between computation and I/O, demonstrates that billion-scale graphs could be trained efficiently without costly distributed setups. This work, DiskGNN [5], was accepted at **SIGMOD 2025** as my first-author publication.

Research Experience at NYU Systems Group

Seeking to broaden my expertise, I joined the NYU Systems Group to work with **Prof. Jinyang Li**, pivoting from GNNs to core bottlenecks in **LLM and AI pipelines**. I started by building robust Retrieval-Augmented Generation (RAG) solutions. I first built PolyG [4], which improves **Graph-based RAG** that augments LLM generation using information retrieved from a knowledge graph. After an extensive study, I observed that existing GraphQA benchmarks were biased towards one type of question on object discovery. As such, I curated a new GraphRAG benchmark, PolyBench, which covers a comprehensive set of question patterns. Moreover, since current methods are not designed for solving diverse question types, I developed a novel GraphRAG solution that adaptively selects suitable graph traversal methods for context retrieval with 4x speedup on end-to-end latency. Both PolyG and PolyBench are fully open-source at <https://github.com/Liu-rj/PolyG>. I plan to submit PolyG to KDD 2026 as the first author.

While developing RAG pipelines, I noticed that the LLM deployment has clear performance bottlenecks. I am currently working on a project to optimize **RAG serving latency** by reusing KV caches across queries. Observing that existing systems are heavily I/O bounded by loading KV caches from disk, I proposed a novel mixed-precision KV compression approach that eliminates the I/O bottleneck while maintaining LLM

generation quality, with a tailored pipeline to overlap KV compression with prefill computation. I plan to submit this work to ICML 2026 as the first author.

Beyond system optimizations, I am also deeply interested in algorithmic developments. When building PolyG, it became clear that information retrieval is just one component of an intelligent agent, and that current agentic frameworks, such as ReAct and Reflexion, are fundamentally brittle. As such, I am now also working on proposing a **new system abstraction for general LLM agents**, using Intermediate Representation (IR) to express agent actions for improved reliability and explainability.

Research Plan: In my future PhD study, I am eager to continue research on system and algorithm optimizations for AI pipelines, particularly for LLM agents. I believe that **versatile, efficient, and robust agent systems** are key to unlocking future AI developments. Specifically, 1) current LLM agents are intrinsically unstable, and thus new system abstractions for more reliable agent execution are required; 2) agent programs are observed to have a rigorous development phase, where the agents are refined by interleaving evaluation and optimization, and thus this stage necessitates robust system support; 3) the deployment of agentic workflows consists of both LLM generation and external function calls, and thus joint optimizations on end-to-end resource utilization and scheduling are needed. This integrated approach—designing better abstractions and optimizing their execution workflows—is the line of research that I am most passionate about and wish to pursue in my PhD. In addition, I am also excited to explore topics broadly in building cutting-edge AI infrastructure and applications.

Why MIT: MIT is my top choice for pursuing this vision. The work of **Prof. Omar Khattab** on self-improving AI pipelines (DSPy) and retrieval models (Colbert) aligns perfectly with my background in RAG and system abstractions. I am also keen to work with **Prof. Adam Belay**, whose recent work on agentic workflows (Murakkab) matches my intended trajectory toward next-generation agent systems. Additionally, **Prof. Samuel Madden**’s innovation in combining database systems and AI presents an exciting avenue for agentic data management. I am also strongly interested in the pioneering work of **Prof. Song Han** on sparse attention and quantization, which focuses the efficient inference challenges I aim to tackle. The collaborative environment at MIT would be invaluable as I work to bridge foundational systems and intelligent applications. I am confident I can make meaningful contributions to the research community at MIT.

References

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